

# How our notebook argues

Jim Simons asked whether daily prices and volumes predict stock returns well enough to trade on. Our answer is no. They rank large stocks barely better than chance, and what  $R^2_{\text{OOS}}$  shows as skill is mostly the average return.

**Does the line of thought hold?** Yes. Each task answers the question the previous one raises, and all six steps arrive at the same answer. The weak points are the ones we state ourselves: a short test period with one crash, and today's index members.

## The argument, step by step

### Task #1

8 lab stocks

#### Do returns and volumes move together?

Volume jumps with the size of a price move (correlation 0.27 to 0.47 with the absolute return), barely with its sign (−0.12 to +0.05), and does not predict the next day's return (−0.05 to +0.03).

**So** volume enters as a signal relative to the stock's own past year, next to past returns.

### Task #2

481 S&P 500 stocks

#### What do OLS and Ridge achieve?

$R^2_{\text{OOS}}$  of 0.30% at 5 days and 0.99% at 20 days, but the average return alone reaches 0.27% and 0.96%: 88% and 97% of it is drift. Ridge barely differs from OLS with half a million observations.

**So**  $R^2_{\text{OOS}}$  is a weak measure of skill. From here on we benchmark with the rank IC (each day's rank correlation between forecasts and returns across stocks, the ranking a portfolio needs) and with the Sharpe ratio of portfolios.

### Task #2 bis

12 signals

#### Do the market and more signals help?

For a linear model the market changes nothing: it adds the same amount to every stock, so the rank IC stays at 0.028 (5 days) and 0.024 (20 days). Six extra stock signals raise the rank IC on the validation days (to 0.026 and 0.064) but cut it on the test days (to 0.006 and 0.008).

**So** we keep all 12 signals, the best set on validation, and ask whether nonlinear models can use the market to rank stocks.

### Task #3

7 models, tuned

#### Which machine learning method performs best?

All settings are chosen by the rank IC on the validation days. On the test days the forest ranks best (rank IC 0.012 and 0.020), XGBoost follows at 20 days (0.015), the linear models stay below 0.01, and NN3 ranks no better than chance. On the validation days NN3 had ranked best at both horizons (0.042 and 0.082).

**So** validation rankings did not carry over, and re-estimating every year does not fix that. NN3's flexibility probably goes into timing the market, which ranks no stock; the forest gains a little from the market signals.

### Task #4

NN3, top and bottom 10

#### Does the ML portfolio make money?

After 1 bp fees, \$100 become \$126 weekly and \$186 monthly in the long portfolio; the shorts lose. Only the monthly long portfolio reaches the edge of chance (Sharpe ratio 0.90, better than 95.1% of the random picks), which one of six strategies does about one time in four, and among the larger stocks that edge disappears. NN3 beats both one-line beta rules in four of six portfolios, but no portfolio earns a significant positive alpha against seven factors.

**So** our best result sits at the edge of luck and comes from a model whose ranking is no better than chance.

### Conclusion

one figure

#### What do we tell Jim Simons?

Ranking skill and portfolio returns often disagree: at 5 days the best rankers build the worst portfolios, and at 20 days only XGBoost's long-short portfolio clears the 95% line of luck, one of 16 tries. Gu et al. find that a network forecasts best and earns alpha; in our setting it does neither. A 2.1-year test cannot separate skill from luck: a Sharpe ratio measured over it has a standard error of about 0.68.

## Questions a grader may ask

### Why the S&P 500 instead of the 8 lab stocks?

Eight stocks are too few for machine learning and long-short portfolios. Task #1 still uses only the lab files, and the Yahoo returns match them (correlation 0.984 to 1.000).

### Why no 60-day horizon?

The 2.2-year test period would hold fewer than nine non-overlapping quarters.

### Why the rank IC and the Sharpe ratio after Task #2?

Task #2 shows that  $R^2_{\text{OOS}}$  lets the average return pass for skill, and RMSE shares its other weaknesses. The rank IC judges only the ranking of stocks, ignores the market's level and weighs every day equally; the Sharpe ratio measures what the ranking earns.

### Why NN3 for the portfolio?

It is chosen like every setting: by the rank IC on the validation days, where NN3 ranks best at both horizons; it is also Gu et al.'s best model. The forest ranks best on the test days, but picking the model by its test results would be hindsight, and at 20 days its long-short portfolio would even have earned less (Sharpe ratio 0.40 against 0.68).

### Why tune on the rank IC?

Settings should be chosen by what we evaluate. Tuned on squared errors, XGBoost used only the two market signals and could not rank at all. A day on which a model gives every stock the same forecast counts as a rank IC of 0, so the tuning cannot favour a model that ranks only on a few days. The cost: one short validation period makes the choice noisy, and the model rated best there ranked worst on the test days at both horizons.

### Why not re-estimate every year, as Gu et al. do?

We tried it (3.5): most models change by at most 0.002, while the forest at 5 days and NN3 lose. The year matters far more than the estimation: on average our models rank well in 2019 and worse than chance in 2018 and 2020.

### What about survivorship bias?

Our universe is today's index. Among the larger half of the stocks the monthly long portfolio's edge disappears (Sharpe ratio 0.40) while the weekly portfolios improve, so the bias may flatter our best result.

## Before we submit

- Read the Conclusion: it answers each question of the slides in one line.
- Upload only `PCLab4_Group4.ipynb`, as for Labs 1–3; all outputs are in the file. Re-running it needs the two cache files next to it (`pclab4_ml_forecasts.npz`, `pclab4_yearly_forecasts.npz`, both in `lab4` on GitHub) and takes about a minute, or about 23 minutes without them.